The Age Dependence of the *Kepler* Radius Valley Is Confined to the First Gigayear:

A Confounder-Controlled Re-analysis with Forward-Modelled Age Uncertainties

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Abstract

The number ratio of super-Earths to sub-Neptunes has been reported to rise with host-star age by several groups and is widely read as evidence for Gyr-long core-powered mass loss. Stellar age, however, is entangled with stellar mass, metallicity and transit detectability in every *Kepler* sample, and isochrone ages carry uncertainties comparable to the intrinsic spread of log age. We re-analyse the super-Earth fraction $f_{\rm SE}$ for 2,007 *Kepler* planets with $1 < R_p < 4\,R_\oplus$ (after scaling to the mass- and period-dependent valley) and $1 < P < 100$ d around 1,510 hosts, combining the *Gaia–Kepler* Stellar Properties Catalog with the CKS VII sample, and introduce three controls that have not been applied to this question: a per-star detectability covariate derived from the detected planet's own multiple-event statistic; a stratified permutation null that shuffles ages only within strata of mass, [Fe/H] and detectability; and an errors-in-variables model that propagates each star's age posterior into a constraint on the sub-Neptune conversion timescale τ. The raw trend is not monotonic: $f_{\rm SE} = 0.31 \pm 0.05$ for hosts younger than 1 Gyr, 0.48 ± 0.03 at 1–2 Gyr and 0.38–0.44 beyond. The old-age decline in the full *Kepler* sample is a sensitivity effect, and the apparent rise in CKS is driven by the age–metallicity anticorrelation; the confounders alone predict a young–old difference of the *opposite* sign (-0.07 ± 0.05 against the observed $+0.12$, $z = 3.7$). After nearest-neighbour matching in mass, [Fe/H], period and detectability, hosts younger than 1 Gyr have $f_{\rm SE} = 0.31$ against 0.47 for matched older hosts ($\Delta = -0.16$, 68% interval $[-0.22, -0.11]$, host-level bootstrap $p = 0.007$), a deficit present independently in the CKS and non-CKS halves, located at 1.0–1.5 $R_\oplus$ and mirrored by an excess just above the valley (1.8–2.3 $R_\oplus$: 24% versus 12%). For hosts older than 2 Gyr the matched difference is -0.02 ± 0.02. The errors-in-variables fit prefers a saturating conversion with $\tau = 0.4$ Gyr (2σ: 0.15–1.5 Gyr; $\Delta\chi^2 = 7.7$ against a static population), while continued log-linear evolution improves on a static population by only $\Delta\chi^2 = 2.7$. A power analysis shows that a rise of $f_{\rm SE}$ from 0.27 to 0.44 between 0.5 and 8 Gyr would have produced a measured-age slope of $+0.66 \pm 0.20$ against $+0.39 \pm 0.21$ observed. The radius distribution therefore does evolve, but the evolution that current data resolve is largely complete within 1–2 Gyr, and the age trend as measured with *Kepler* isochrone ages does not discriminate between core-powered mass loss and photoevaporation.

Keywords: exoplanets — planetary atmospheres — stellar ages — methods: statistical

1. INTRODUCTION

The radius distribution of close-in planets smaller than Neptune is bimodal, with a valley near $1.8\,R_\oplus$ separating rocky super-Earths from sub-Neptunes that retain H/He envelopes (Fulton and Petigura 2018; Fulton et al. 2017; Van Eylen et al. 2018). Two families of models carve the valley by stripping envelopes: photoevaporation by the host's X-ray and ultraviolet flux, which is strongest in the first ∼100 Myr (Lopez and Fortney 2013; Owen and Wu 2013, 2017), and core-powered mass loss, which is driven by the planet's own cooling luminosity and proceeds over gigayears (Ginzburg et al. 2018; Gupta and Schlichting 2019). The two make a differential prediction that is in principle easy to test: under photoevaporation the relative abundance of super-Earths should be set within ∼100–300 Myr, whereas under core-powered mass loss it should keep rising, "especially over periods from 500 Myr to 3 Gyr" (Gupta and Schlichting 2020).

Several groups have reported the predicted rise. Berger et al. (2020a) found a super-Earth to sub-Neptune number ratio of 0.61 ± 0.09 for *Kepler* hosts younger than 1 Gyr against 1.00 ± 0.10 for older hosts, using *Gaia*-based isochrone ages. David et al. (2021) found the valley emptier below ∼2–3 Gyr in the California–*Kepler* Survey (CKS) sample and a deficit of large super-Earths (1.5–$1.8\,R_\oplus$) around young hosts; Sandoval et al. (2021) recovered the rise with a Monte-Carlo treatment of age uncertainties. Chen et al. (2022) found the ratio increasing with age and decreasing with [Fe/H] and [α/Fe] in a LAMOST–*Gaia–Kepler* sample, Ho and Van Eylen (2023) a weak positive dependence of the valley location on age, and two 2026 studies report the same qualitative behaviour in a homogeneous spectroscopic host sample (Kamulali et al. 2026) and in a sample with chemical and kinematic ages (Chen et al. 2026). The result is now routinely cited as evidence for Gyr-long atmospheric loss.

The age axis is, however, unusually exposed to confounding. First, isochrone ages are a function of position in the Hertzsprung–Russell diagram and hence of mass: a dwarf receives a young age because it sits near the zero-age main sequence, and informative ages exist mainly for stars massive enough to evolve within a Hubble time. In the full *Kepler* sample the median host mass falls from $1.28\,M_\odot$ in the < 1 Gyr bin to $0.91\,M_\odot$ above 10 Gyr, and the valley location, the sub-Neptune size and the super-Earth/sub-Neptune ratio all depend on mass (Berger et al. 2020a; Fulton and Petigura 2018; Petigura et al. 2022). Second, age anticorrelates with metallicity, and metal-poor hosts have fewer sub-Neptunes (Owen and Murray-Clay 2018; Petigura et al. 2018). Third, old hosts are harder targets: in the full *Kepler* sample the oldest bins are fainter by 1.8 mag in K_p, so the small planets that define the super-Earth population are preferentially lost, which biases the ratio *downward* with age. Petigura et al. (2022) found no change of sub-Neptune size with age over 1–10 Gyr and warned explicitly that metallicity and age trends are confounded with mass. Previous analyses addressed these issues by matching the mass distributions of two age bins (Berger et al. 2020a) or by treating [Fe/H] as a second axis (Chen et al. 2022); none quantified what the confounders alone would produce, none controlled detectability star by star, and none propagated the $\sim$0.2–0.4 dex age uncertainties into a model of the evolution itself.

Here we do those three things. We ask the question the way the physics poses it—not "is there a trend" but "on what timescale does the population change, and could the data see a Gyr-scale change if it were there". Section 2 describes the samples, Section 3 the tests, Section 4 the results, Section 5 a cross-check with stellar rotation, and Section 6 the implications.

2. DATA

2.1. Catalogs and samples

Three overlapping samples are analysed (Table 1); sample C is the primary one. Sample A combines the *Kepler* Data Release 25 (DR25) Kepler Objects of Interest (KOIs; Thompson et al. 2018) with the *Gaia–Kepler* Stellar Properties Catalog (GKSPC; Berger et al. 2020b), whose isochrone masses, radii, metallicities and ages derive from *Gaia* DR2 parallaxes, 2MASS K photometry and spectroscopic or photometric [Fe/H]. Sample B is the CKS VII sample (Fulton and Petigura 2018): Keck-HIRES spectroscopic $T_{\rm eff}$ and [Fe/H] (Johnson et al. 2017), *Gaia* DR2 radii and grid-based isochrone ages. Sample C takes the CKS ages and metallicities where available and adds the sample-A planets whose hosts are not in CKS, so that every host appears once with its best available age. All catalogs were retrieved from public GitHub mirrors of the original tables.[1]

[1]GKSPC Tables 1–2 and the DR25 KOI list from https://github.com/ExoJulia/SysSimData and https://github.com/ExoJulia/SysSimDataInputs (Ford et al. 2020; Hsu et al. 2019); the CKS VII machine-readable tables from https://github.com/California-Planet-Search/cksgaia; rotation periods of Mazeh et al. (2015) matched to CKS from https://github.com/RuthAngus/kepler_ages.

Table 1. Samples analysed. Planet counts are after the radius and period cuts of Section 2.4.

Sample	Planets	Hosts	Median σ(log age)
A: DR25 KOIs × GKSPC	1,759	1,350	0.37 dex
B: CKS VII	1,019	639	0.19 dex
C: B + non-CKS part of A	2,007	1,510	0.29 dex

2.2. Cuts

For sample A we follow Berger et al. (2020a): DR25 pipeline disposition CANDIDATE and no false-positive disposition; *Gaia* RUWE ≤ 1.2; isochrone goodness-of-fit ≥ 0.99; main-sequence hosts ($\log g > 3.8$, no red-giant or clump flag, $T_{\rm eff} < 7900$ K); and informative ages only (time to terminal-age main sequence < 20 Gyr; no GKSPC unreliable-age flag). Planet radii are recomputed as $R_p = (R_p/R_\star)_{\rm DR25}\,R_{\star,\rm GKSPC}$. For sample B we require no CKS or archive false-positive flag, a *Gaia* $8''$ flux ratio below 1.1 (no dilution) and $R_\star < 2\,R_\odot$. Ages and uncertainties are the catalog posterior medians and 16th/84th percentiles, used in log age with $\sigma = \max(\sigma_+, \sigma_-)$.

In sample A the variance of measured log age (0.077 dex^2) is smaller than the mean squared age uncertainty (0.138 dex^2); in sample C the two are 0.088 and 0.121 dex^2. Measured ages therefore order the hosts only coarsely beyond $\sim$2 Gyr, which is why Section 4.6 quantifies the power of every test before a null is interpreted.

2.3. Detectability proxy

For each planet the DR25 multiple-event statistic (MES) is rescaled to the value a $1\,R_\oplus$ planet would have produced at the same period around the same star,

$$\mathrm{MES}_1 = \mathrm{MES}\,(R_\oplus/R_p)^2. \tag{1}$$

To first order MES_1 depends only on the star's noise, radius and transit count, not on the detected planet's radius, so $\log \mathrm{MES}_1$ is a per-star sensitivity measure that can be used as a regression covariate and as a matching variable without a completeness model. The DR25 detection probability $p_{\rm det}(\mathrm{MES}) = 0.94\,\Gamma(30.87, \mathrm{MES}/0.271)$, the Christiansen (2017) parametrisation adopted by Hsu et al. (2019), defines a detection-complete subsample with $p_{\rm det}(1\,R_\oplus) > 0.9$.

2.4. The radius valley and the classification

Radii are scaled along the valley to $P = 10$ d and $M_\star = 1\,M_\odot$,

$$R' = R_p\,(P/10\,\mathrm{d})^{0.10}\,(M_\star/M_\odot)^{-0.25}, \tag{2}$$

with the period and mass slopes of Petigura et al. (2022) and Ho and Van Eylen (2023). The valley is located at the kernel-density minimum of $\log R'$ in each full sample, $1.82\,R_\oplus$ (A) and $1.83\,R_\oplus$ (B); planets with $1 < R' < 1.82\,R_\oplus$ are super-Earths and $1.82 < R' < 4\,R_\oplus$ sub-Neptunes, and only $1 < P < 100$ d is used. Section 4.3 repeats the key test with a fixed $1.8\,R_\oplus$ boundary, period-only scaling, a mass slope of 0.10 and boundaries of 1.7–2.0 $R_\oplus$.

3. METHODS

The statistic throughout is the super-Earth fraction $f_{\rm SE} = N_{\rm SE}/(N_{\rm SE}+N_{\rm SN})$ among detected planets (the ratio used by earlier work is $f_{\rm SE}/(1-f_{\rm SE})$). The covariate vector is $\mathbf{x} = (\log M_\star, [{\rm Fe/H}], \log P, \log {\rm MES}_1)$.

Binned fractions. $f_{\rm SE}$ in six log-age bins (edges 8.5, 9.0, 9.3, 9.6, 9.8, 10.0, 10.3) with bootstrap 68% intervals.

Logistic regressions. $\mathrm{logit}\,P({\rm SE}) = a + b_{\rm age}\log({\rm age}) + \mathbf{b}\cdot\mathbf{x}$, adding covariates one at a time, with standard errors cluster-robust by host star because multi-transiting systems contribute correlated planets.

Stratified permutation null. Hosts are placed in 36 strata (quartiles of $\log M_\star$ × terciles of [Fe/H] × terciles of $\log {\rm MES}_1$) and the measured ages are shuffled *within* strata 4,000 times. This destroys any dependence of the planet population on age while preserving the joint distribution of age with mass, metallicity and sensitivity. The null distributions of the age-only logistic slope and of $f_{\rm SE}(>1\,{\rm Gyr}) - f_{\rm SE}(<1\,{\rm Gyr})$ are compared with the observed values; z is quoted relative to the null mean and spread. An unstratified shuffle is run alongside.

Matched comparison. Each planet around a host younger than a cut (1, 2 or 3.2 Gyr) is matched to its $k = 5$ nearest older planets in standardised $\mathbf{x}$ space, and $f_{\rm SE}({\rm young}) - f_{\rm SE}({\rm matched\ old})$ is bootstrapped over young *host stars* (3,000 resamples). Variants use $k = 1$ and 10 and add $\log R_\star$ and K_p to the matching space.

Errors-in-variables evolution model.

$$\mathrm{logit}\,P({\rm SE}\,|\,t,\mathbf{x}) = a + \mathbf{b}\cdot\mathbf{x} + D\,g(t;\tau), \qquad (3)$$

with $g = 1 - e^{-t/\tau}$, a saturating conversion of sub-Neptunes into super-Earths with timescale τ and asymptotic logit shift D, bounded to $|D| \le 3$ (odds ratios up to $e^3 \approx 20$); and, as the opposite limiting case, $g = \log_{10}(t/{\rm Gyr})$. Each star's true age is integrated out numerically with $K = 128$ draws from a log-normal approximation of its posterior truncated to 0.05–14 Gyr. The likelihood is profiled over τ on a grid from 0.15 to 20 Gyr; $\Delta\chi^2$ is quoted against the static model $D = 0$.

Power analysis. Synthetic super-Earth labels are drawn from the fitted covariate model plus an injected evolution (D,τ), with true ages drawn from the stars' posteriors, and the naive measured-age slope (with covariates) and the errors-in-variables $\Delta\chi^2$ are recomputed over 120 realisations per scenario.

4. RESULTS

4.1. *The raw trend is not monotonic*

In all three samples $f_{\rm SE}$ is lowest for hosts younger than 1 Gyr and highest at 1–2 Gyr; beyond 2 Gyr it declines in the full *Kepler* sample and is flat in CKS (Figure 1, Table 2). The lower panels of Figure 1 show what moves along the age axis: in sample A the median host mass falls from 1.28 to 0.91 $M_\odot$ and the sensitivity to a 1 $R_\oplus$ planet drops by 0.3 dex; in CKS the median [Fe/H] falls from +0.14 to −0.23.

The Berger et al. (2020a) split reproduces in sample A as SE/SN = 0.56 (< 1 Gyr) against 0.71 (> 1 Gyr), but the > 1 Gyr value is an average over a bin-to-bin *decline* from 1.0 at 1–2 Gyr to 0.45 above 10 Gyr. The David et al. (2021) rise reproduces in sample B only between the < 1 Gyr bin and ∼2 Gyr.

4.2. *Confounders explain the old-age behaviour and predict the opposite sign*

Adding mass and metallicity to the logistic model barely moves the age coefficient; adding the per-star detectability term does (Table 3, Figure 3). In sample A the age-only coefficient is -0.51 ± 0.18—$f_{\rm SE}$ *falls* with measured age, $p = 0.006$—and becomes $+0.25 \pm 0.28$ with $\log {\rm MES}_1$. Within detectability tertiles the decline largely disappears (Figure 2): the low-sensitivity tertile has $f_{\rm SE} \approx 0.13$ at every age because only sub-Neptunes are detectable there, and the high-sensitivity tertile has $f_{\rm SE} \approx 0.6$–0.7. In sample B the raw coefficient of $+0.25 \pm 0.22$ drops to +0.06 when [Fe/H] is added and recovers to $+0.43 \pm 0.30$ with detectability. The full-covariate coefficient in sample C is $+0.39 \pm 0.21$ ($p = 0.07$).

The stratified permutation null makes the direction of the confounding explicit (Figure 4). Shuffling ages within mass × [Fe/H] × sensitivity strata, the expected $f_{\rm SE}(>1\,{\rm Gyr}) - f_{\rm SE}(<1\,{\rm Gyr})$ is -0.08 ± 0.06 (A), -0.04 ± 0.06 (B) and -0.07 ± 0.05 (C): the host population alone would make *old* hosts look poorer in super-Earths. The observed differences are +0.06, +0.14 and +0.12, i.e. $z = +2.3$, +2.8 and +3.7 above the confounder-only expectation. The age-only slope tells the same story with less power ($z = +1.5$, +2.7, +2.1). An unstratified shuffle gives the familiar but misleading answer that sample A's negative slope is "significant" ($p = 0.005$) and sample B's positive slope is not.

4.3. *A robust deficit of super-Earths around hosts younger than ∼1 Gyr*

Matching each young planet to its five nearest older planets in $(\log M_\star, [{\rm Fe/H}], \log P, \log {\rm MES}_1)$ gives the cleanest number. In sample C, hosts younger than 1 Gyr (78 planets, 55 stars; median $M_\star = 1.00\,M_\odot$, [Fe/H] = +0.11, $T_{\rm eff} = 5600$ K, $K_p = 13.7$) have $f_{\rm SE} = 0.31$ against 0.47 for their matched older counterparts: $\Delta = -0.16$, 68% interval $[-0.22, -0.11]$, host-level bootstrap $p = 0.007$. The covariate means of the young and matched sets agree to within 0.01–0.02 in every dimension. With the cut at 2 Gyr the difference is -0.05 $[-0.08, -0.03]$ ($p = 0.04$); at 3.2 Gyr it is -0.02 $[-0.04, -0.01]$ ($p = 0.16$).

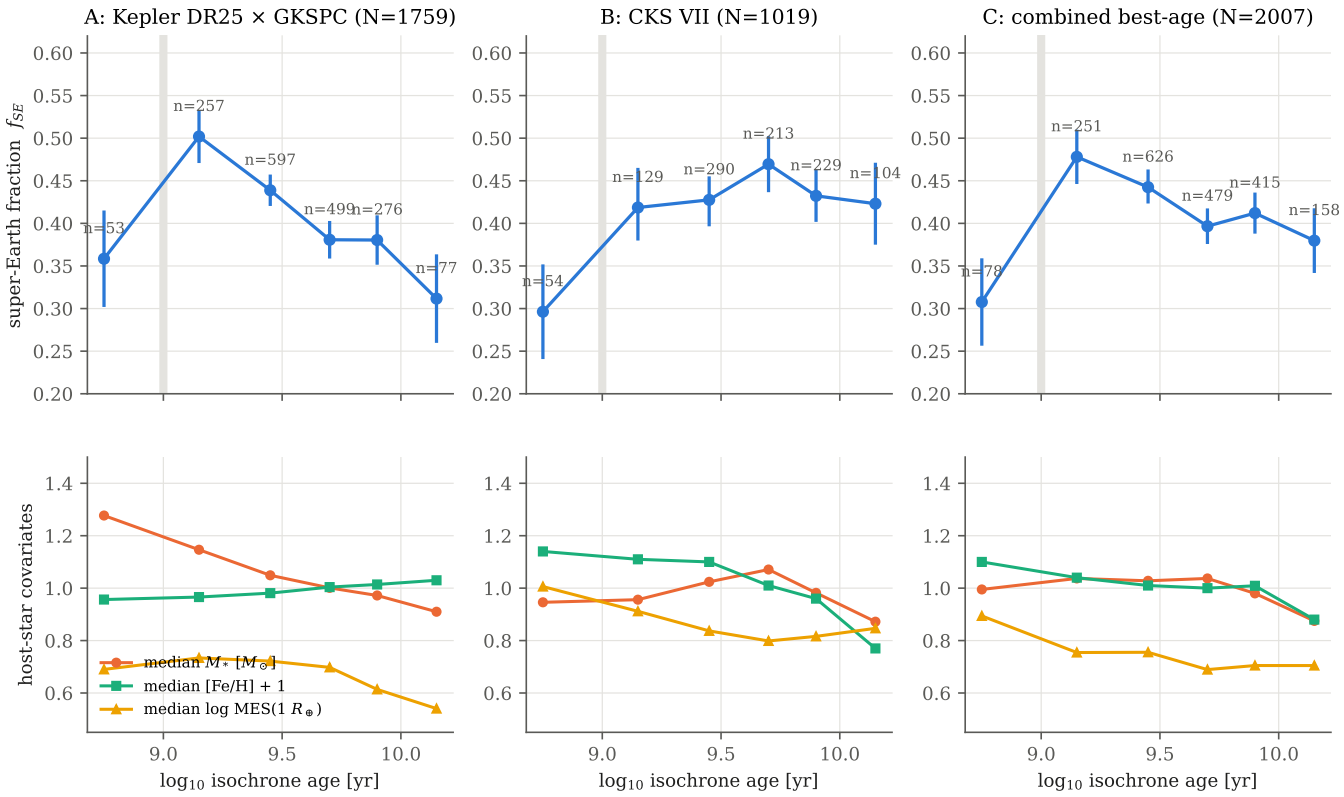


Figure 1. Top: super-Earth fraction in log-age bins for the three samples, with bootstrap 68% intervals and bin counts; the grey band marks 1 Gyr. Bottom: median host mass, [Fe/H] and $\log \mathrm{MES}_1$ in the same bins, showing the demographic and sensitivity gradients that run along the age axis.

Table 2. Super-Earth fraction in log-age bins (bootstrap 68% uncertainties; planet counts in parentheses).

Sample	<1 Gyr	1–2 Gyr	2–4 Gyr	4–6.3 Gyr	6.3–10 Gyr	>10 Gyr
A	0.36 ± 0.06 (53)	0.50 ± 0.03 (257)	0.44 ± 0.02 (597)	0.38 ± 0.02 (499)	0.38 ± 0.03 (276)	0.31 ± 0.05 (77)
B	0.30 ± 0.06 (54)	0.42 ± 0.04 (129)	0.43 ± 0.03 (290)	0.47 ± 0.03 (213)	0.43 ± 0.03 (229)	0.42 ± 0.05 (104)
C	0.31 ± 0.05 (78)	0.48 ± 0.03 (251)	0.44 ± 0.02 (626)	0.40 ± 0.02 (479)	0.41 ± 0.02 (415)	0.38 ± 0.04 (158)

The deficit is present in the two disjoint halves of sample C—the CKS hosts (-0.17, $p = 0.02$) and the non-CKS *Kepler* hosts (-0.13, $p = 0.09$)—whose ages come from different spectroscopic inputs and isochrone pipelines.

Table 4 shows the robustness grid. The sign and size are stable; the subsample p-values track the number of young planets. The deficit is not a hot-star or high-mass effect (it is the same below 6300 K and below $1.2\,M_\odot$), not a vetting effect (it persists at MES > 20, where the DR25 Robovetter completeness is high) and not a boundary artefact.

4.4. *Where in radius the change happens*

The young hosts' planets are missing at the small end and piled up just above the valley (Figure 5). Fractions of planets by scaled radius, young versus matched old, are: 1.0–1.5 $R_\oplus$, 0.21 versus 0.34 ($\Delta = -0.14$ $[-0.19, -0.09]$); 1.5–1.82 $R_\oplus$, 0.10 versus 0.13; 1.82–2.3 $R_\oplus$, 0.24 versus 0.12 ($\Delta = +0.12$ $[+0.07, +0.18]$, $p = 0.02$); 2.3–3 $R_\oplus$, 0.28 versus 0.27; 3–4 $R_\oplus$, 0.17 versus 0.13. The sub-Neptune peak of the young sample is at $\sim$2.4 $R_\oplus$ rather than $\sim$2.7 $R_\oplus$. This is the pattern expected if a population of thin-envelope planets sitting just above the valley has not yet been stripped at <1 Gyr and ends up below it later; it differs from David et al. (2021), who located the young deficit at 1.5–1.8 $R_\oplus$.

4.5. *Timescale of the change*

Propagating the age posteriors, the saturating model is preferred over a static population with $\Delta\chi^2 = 7.7$ (two extra parameters, $p = 0.02$) in sample C, 4.6 in A and 4.8 in B (Figure 6, Table 5). The best timescale is $\tau = 0.38$ Gyr in C (0.30 in A, 0.48 in B); the 2σ interval in C is 0.15–1.5 Gyr, and $\tau = 3$ Gyr is worse than the best fit by $\Delta\chi^2 = 5.2$. The implied intrinsic f_{SE} at mean covariates rises from 0.23 at 0.5 Gyr to 0.35 at 1 Gyr and 0.40 by 2 Gyr, then stays flat. The log-linear alternative—a continued rise of 0.33 in logit per dex, i.e. f_{SE} from 0.30 to 0.41 between 0.3 and 8 Gyr—improves on the static model by only

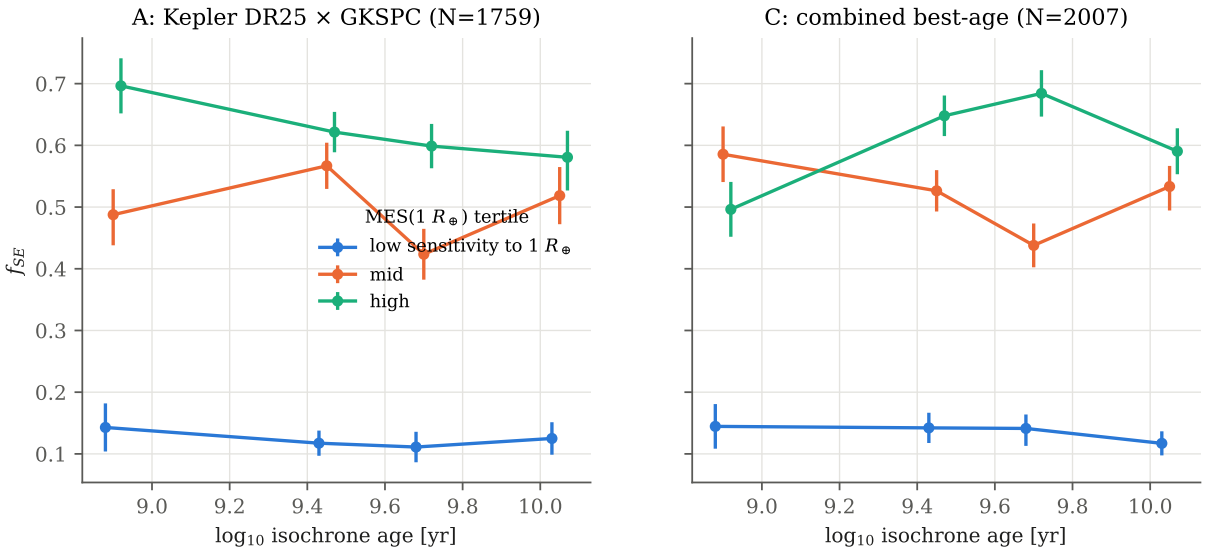


Figure 2. Super-Earth fraction versus age inside tertiles of $\log \mathrm{MES}_1$ for samples A and C. Sensitivity, not age, sets most of the old-age decline of Figure 1.

Table 3. Logistic coefficient on log age, b_{age} (cluster-robust standard errors), as covariates are added, and the stratified-permutation results.

Model	A	B	C
age only	-0.51 ± 0.18	$+0.25 \pm 0.22$	-0.16 ± 0.16
+ $\log M_\star$	-0.51 ± 0.21	$+0.31 \pm 0.22$	-0.12 ± 0.17
+ [Fe/H]	-0.51 ± 0.22	$+0.06 \pm 0.26$	-0.16 ± 0.17
+ $\log P$	-0.48 ± 0.26	-0.07 ± 0.29	-0.13 ± 0.20
+ $\log \mathrm{MES}_1$	$+0.25 \pm 0.28$	$+0.43 \pm 0.30$	$+0.39 \pm 0.21$
Stratified null, $f_{\mathrm{SE}}(>1\,\mathrm{Gyr}) - f_{\mathrm{SE}}(<1\,\mathrm{Gyr})$:			
observed	$+0.06$	$+0.14$	$+0.12$
null mean $\pm$ sd	-0.08 ± 0.06	-0.04 ± 0.06	-0.07 ± 0.05
z	$+2.3$	$+2.8$	$+3.7$
Stratified null, age-only slope:			
observed	-0.51	$+0.25$	-0.16
null mean $\pm$ sd	-0.70 ± 0.13	-0.15 ± 0.15	-0.42 ± 0.12
z	$+1.5$	$+2.7$	$+2.1$

$\Delta\chi^2 = 2.7$ ($p = 0.10$) and is worse than the saturating model by 5.0 with one fewer parameter. The amplitude D sits at its bound of 3 in samples A and C, so the 1σ ranges in τ are not meaningful; the 2σ ranges and the comparisons at fixed τ are.

4.6. *Could the data have seen a Gyr-scale change?*

Yes, for a change of the size claimed in the literature (Figure 7, Table 6). With sample C's own hosts, age posteriors and covariates, a static population yields a naive measured-age slope of 0.00 ± 0.20, and an injected core-powered-like evolution with $D = 1$, $\tau = 3$ Gyr (f_{SE} 0.27 $\rightarrow$ 0.33 $\rightarrow$ 0.44 at 0.5, 2 and 8 Gyr) yields $+0.66 \pm 0.20$ and an errors-in-variables $\Delta\chi^2 > 6$ in 88% of realisations. The observed full-covariate slope, $+0.39 \pm 0.21$, lies between the static and the

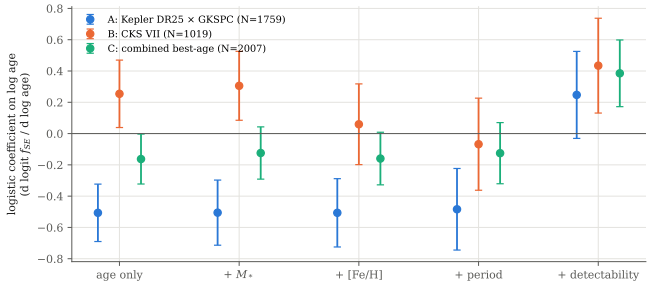


Figure 3. Logistic coefficient on log age as covariates are added, with cluster-robust errors (Table 3).

$D = 1$, $\tau = 1$ Gyr scenario ($+0.37 \pm 0.18$) and 1.3σ below the $\tau = 3$ Gyr scenario; a stronger late evolution ($D = 2$, $\tau = 3$ Gyr, f_{SE} 0.17 $\rightarrow$ 0.50) would have given $+1.2 \pm 0.2$ and is excluded. Mild continued evolution ($D = 0.6$, $\tau = 3$–8 Gyr; expected $+0.25$ to $+0.39$) is fully consistent with the data: the measurement cannot yet separate "finished by 1–2 Gyr" from "mostly finished by 2 Gyr with a slow tail".

5. Cross-check with stellar rotation

Rotation periods from Mazeh et al. (2015) exist for 217 of the CKS hosts with $T_{\mathrm{eff}} < 6250$ K. Rotation is an age indicator whose systematics are unrelated to the isochrone age–mass degeneracy, so it can test whether the isochrone-young hosts are physically young and whether the deficit follows an independent age proxy. The isochrone ages are physically meaningful: the rotation residual $\Delta \log P_{\mathrm{rot}}$ (log period minus a quadratic fit in T_{eff}) correlates with isochrone log age at $\rho = 0.41$ ($p = 4 \times 10^{-10}$), and the isochrone-young hosts rotate faster (median $\Delta \log P_{\mathrm{rot}} = -0.10$ against $+0.03$). On the

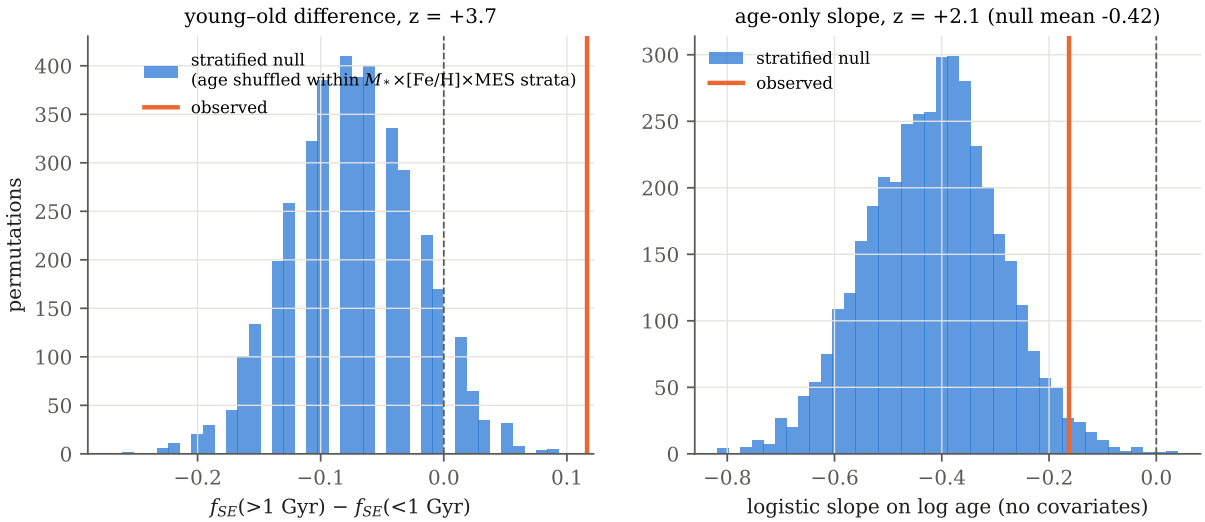


Figure 4. Sample C: stratified-null distributions of the young–old difference (left) and of the age-only logistic slope (right), with the observed values. The confounders alone predict the opposite sign.

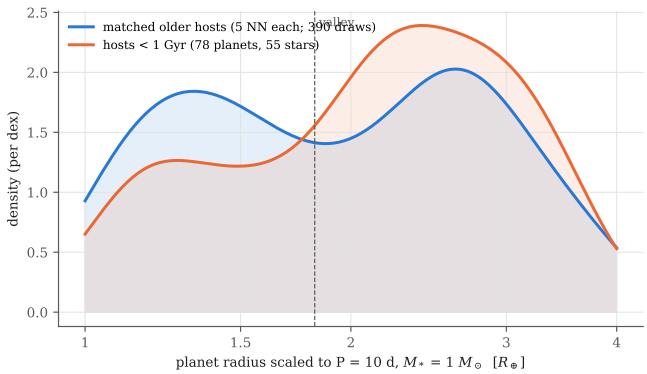


Figure 5. Scaled-radius distributions (kernel density, per dex) of planets around hosts younger than 1 Gyr and around their matched older counterparts, sample C. The dashed line marks the valley at $1.82\,R_\oplus$.

rotation axis itself $f_{\rm SE}$ is flat: 0.44, 0.36 and 0.40 in the fast, intermediate and slow terciles, and the fastest-rotating quartile differs from its matched slow counterpart by $+0.03 \pm 0.06$. This null has no power against the isochrone result: the fastest quartile contains only 20% isochrone-young (< 1 Gyr) hosts against 7% in the rest, so the isochrone-based deficit predicts a fast–slow difference of only -0.03, consistent with $+0.03 \pm 0.07$ observed. Among the 22 isochrone-young hosts that have rotation periods, $f_{\rm SE} = 0.32$, the same as in the full young sample. A decisive rotation-based test needs rotation periods for a few hundred hosts younger than 1 Gyr, i.e. the *TESS* young-host samples rather than *Kepler*.

6. Discussion

6.1. *Does the radius distribution change with age?*

Yes, and the change is not an artefact of host-star demographics: the confounders present in *Kepler* samples push the apparent trend the other way, so the genuine young–old difference ($\approx +0.14$ to $+0.19$ in $f_{\rm SE}$) is larger than the raw one. But the change that the data resolve is a deficit of super-Earths, with an excess of planets just above the valley, around hosts younger than $\sim$1 Gyr. Between 2 and 14 Gyr the matched difference is -0.02 ± 0.02, and the errors-in-variables fit places the conversion timescale at $\tau \approx 0.4$ Gyr with a 2σ upper bound of 1.5 Gyr under a saturating model. The "Gyr-long" character of the published trend comes from averaging a 1–2 Gyr peak with lower, detection-biased or metallicity-biased values at old ages, not from a measured rise beyond 2 Gyr.

6.2. *Implications for the mass-loss mechanism*

Photoevaporation models place the super-Earth population after $\sim$100–300 Myr but with a tail: planets with marginal envelopes, and envelopes that keep contracting, continue to cross the valley for up to $\sim$1 Gyr (Owen and Wu 2017; Rogers and Owen 2021). Core-powered mass loss predicts the ratio to keep rising especially from 0.5 to 3 Gyr (Gupta and Schlichting 2020), and the two mechanisms hand over to each other in different parts of the mass–flux plane (Owen and Schlichting 2024). A conversion largely complete by 1–2 Gyr sits at the boundary of both predictions, and the excess at 1.8–2.3 $R_\oplus$ at < 1 Gyr is the natural signature of the last planets to be stripped under either mechanism. The data therefore do not discriminate between the mechanisms, contrary to the usual reading of the age trend. What is excluded is a Gyr-scale continued rise of the size that would make the discrimination ($f_{\rm SE}$ increasing by $\gtrsim 0.15$ between 2 and 8 Gyr); a rise of $\lesssim 0.08$ is not.

Table 4. Matched comparison of hosts younger than 1 Gyr with their five nearest older hosts in $(\log M_\star, [\mathrm{Fe/H}], \log P, \log \mathrm{MES}_1)$, sample C, under variations of the sample, matching and classification. $\Delta = f_{\mathrm{SE}}(\mathrm{young}) - f_{\mathrm{SE}}(\mathrm{matched\ old})$ with 68% host-bootstrap interval and two-sided bootstrap p.

Variant	N	f_{SE} young	f_{SE} matched old	Δ [68%]	p
baseline ($k = 5$)	2007	0.31	0.47	-0.16 $[-0.22, -0.11]$	0.007–0.013
$k = 1$ / $k = 10$	2007	0.31	0.46 / 0.47	-0.15 / -0.16	0.047 / 0.005
match also on $\log R_\star$, K_p	2007	0.31	0.43	-0.13 $[-0.19, -0.07]$	0.047
MES > 12 (vetting-complete)	1501	0.25	0.39	-0.14 $[-0.21, -0.09]$	0.020
MES > 20	952	0.19	0.32	-0.13 $[-0.18, -0.08]$	0.008
$p_{\mathrm{det}}(1\,R_\oplus) > 0.5$	493	0.39	0.57	-0.18 $[-0.31, -0.07]$	0.105
exclude 3–4 $R_\oplus$	1668	0.37	0.56	-0.19 $[-0.27, -0.12]$	0.013
$P < 30$ d	1640	0.37	0.53	-0.17 $[-0.24, -0.11]$	0.017
fixed 1.8 $R_\oplus$ boundary, unscaled radii	1989	0.28	0.44	-0.16 $[-0.22, -0.10]$	0.004
period-only scaling / mass slope 0.10	1979 / 1992	0.31	0.47 / 0.46	-0.16 / -0.15	0.001 / 0.007
valley at 1.7 / 1.9 / 2.0 $R_\oplus$	2007	0.29 / 0.35 / 0.38	0.42 / 0.48 / 0.51	-0.13 / -0.14 / -0.12	0.02–0.03
$T_{\mathrm{eff}} < 6300$ K / ≥ 6300 K	1817 / 190	0.29 / 0.35	0.47 / 0.49	-0.18 / -0.14	0.02 / 0.19
$M_\star < 1.2$ / $\geq 1.2\,M_\odot$	1732 / 275	0.30 / 0.36	0.45 / 0.53	-0.16 / -0.17	0.03 / 0.23
single / multi-transiting hosts	1159 / 848	0.30 / 0.32	0.49 / 0.42	-0.20 / -0.11	0.015 / 0.21
CKS-only / non-CKS hosts	1019 / 988	0.30 / 0.33	0.47 / 0.47	-0.17 / -0.13	0.024 / 0.092

Table 5. Errors-in-variables fits. $\Delta\chi^2$ is the improvement over a static population ($D = 0$); the saturating model adds two parameters, the log-linear model one.

	A	B	C
best τ (Gyr)	0.30	0.48	0.38
2σ range of τ (Gyr)	0.15–4.9	0.15–7.9	0.15–1.5
$\Delta\chi^2$, saturating (best τ)	4.6	4.8	7.7
$\Delta\chi^2$ at $\tau = 1$ Gyr	1.8	4.4	5.1
$\Delta\chi^2$ at $\tau = 3$ Gyr	0.7	2.1	2.4
$\Delta\chi^2$ at $\tau = 8$ Gyr	0.5	0.9	1.4
$\Delta\chi^2$, log-linear	1.0	2.0	2.7
implied f_{SE} at 0.5 Gyr	0.26	0.22	0.23
implied f_{SE} at 1 Gyr	0.36	0.34	0.35
implied f_{SE} at 2 Gyr	0.39	0.42	0.40
implied f_{SE} at 8 Gyr	0.39	0.43	0.40

Table 6. Power analysis on sample C: measured-age logistic slope (with covariates) and the fraction of realisations with errors-in-variables $\Delta\chi^2 > 6$ under injected evolution scenarios. The observed slope is $+0.39 \pm 0.21$.

D	τ (Gyr)	$f_{\mathrm{SE}}(0.5)$	$f_{\mathrm{SE}}(8)$	slope	$P(\Delta\chi^2 > 6)$
0	–	0.38	0.38	0.00 ± 0.20	0.04
1	0.3	0.34	0.38	$+0.14 \pm 0.21$	0.12
1	1	0.27	0.41	$+0.37 \pm 0.18$	0.40
1	3	0.27	0.44	$+0.66 \pm 0.20$	0.88
1	8	0.30	0.43	$+0.52 \pm 0.18$	0.72
2	0.4	0.27	0.40	$+0.33 \pm 0.18$	0.56
2	3	0.17	0.50	$+1.22 \pm 0.21$	1.00
0.6	3	0.31	0.42	$+0.39 \pm 0.20$	0.40

6.3. Why earlier analyses saw a monotonic rise

In GKSPC-based work, matching the mass distributions of two age bins leaves the brightness and sensitivity differences in place (replacing the mass-scaled valley by a fixed 1.8 $R_\oplus$ boundary changes the young–old contrast by only $\sim$0.01 in our sample, so the classification is not the issue). In CKS-based work the metal-poor old hosts contribute fewer sub-Neptunes. In both, the < 1 Gyr bin carries most of the signal while the axis beyond 2 Gyr carries little information: the intrinsic spread of measured log age is comparable to the age uncertainties, which compresses any late trend toward zero and makes bin-to-bin comparisons at 3–10 Gyr mostly noise and demographics.

6.4. Caveats

(i) The young sample is small and its ages are soft: 55 host stars carry the main result, with isochrone ages of 0.46–0.98 Gyr and $\sigma \approx 0.45$ dex, so their true ages plausibly span 0.2–2.5 Gyr; "< 1 Gyr" means "near the zero-age main sequence for its mass and metallicity". The obvious systematics that would mislabel such stars push planet radii the wrong way to mimic a super-Earth deficit (a hotter T_{eff} makes $R_\star$ and R_p smaller, not larger). (ii) Young stars are active; unocculted spots bias transit depths by a few per cent and activity raises the noise. The MES_1 covariate captures the noise but not spot bias or Robovetter behaviour on variable stars; the deficit survives at MES > 20, which limits but does not eliminate this channel. (iii) Samples A and B share stars and are not independent confirmations; the CKS and non-CKS halves of sample C are, and give -0.17 and -0.13. (iv) MES_1 inherits the Eddington-type upward fluctuation of detected signals near threshold; this affects super-Earths more than sub-Neptunes but is not age dependent, and the MES cuts give the same answer. (v) Age posteriors are approximated as truncated log-normals; asymmetric or multimodal posteriors and the GKSPC grid edge at 14–20 Gyr are not modelled, and D reaches its prior bound in two samples. (vi) The saturating $g(t)$ is a summary of conversion histories that in reality depend on core mass, envelope fraction and flux. (vii) The valley

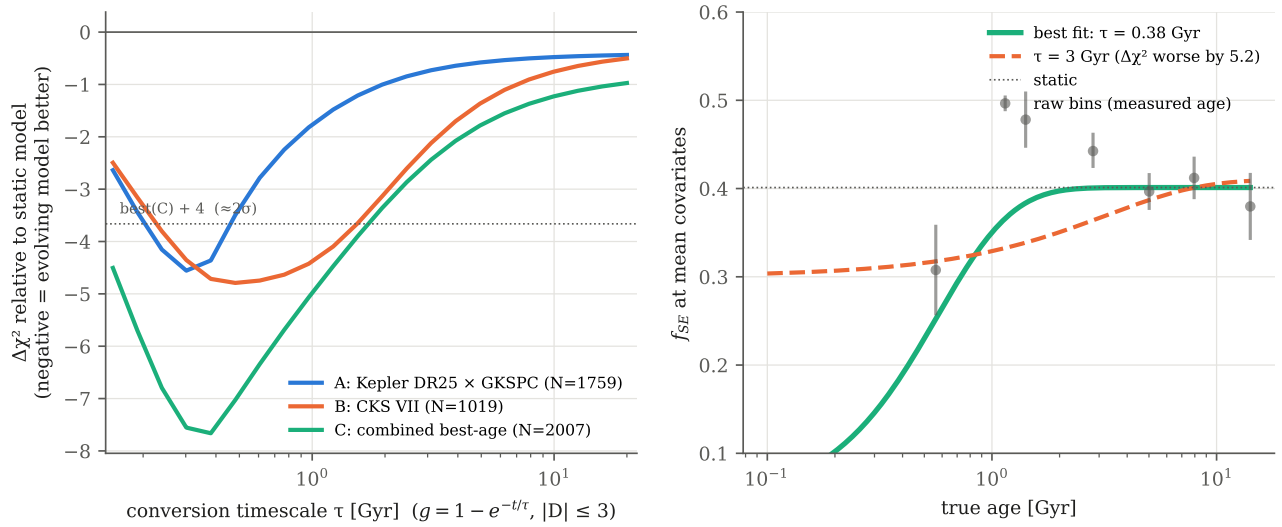


Figure 6. Left: profile likelihood over the conversion timescale τ for the three samples, with $|D| \leq 3$; the dotted line is the sample-C minimum plus 4 ($\approx 2\sigma$). Right: implied intrinsic $f_{\rm SE}(t)$ at mean covariates for the best sample-C fit, for $\tau = 3$ Gyr, and the static model, with the raw measured-age bins overlaid.

slopes are uncertain by $\sim$0.03 (period) and $\sim$0.1 (mass); the robustness grid covers that range. (viii) No kinematic or asteroseismic ages are used; a stratified re-analysis of the Chen et al. (2022) kinematic-age sample with the detectability covariate is the obvious next step.

6.5. *Predictions*

If the conversion saturates by 1–2 Gyr, *TESS* young-host samples with gyrochronological or cluster ages should show $f_{\rm SE}$ rising steeply between 0.2 and 1 Gyr and little change thereafter, with the valley shallowest at a few hundred Myr and the excess just above it. If the log-linear tail is real, PLATO's asteroseismic ages ($\sigma \approx 10\%$) for a few thousand hosts would detect the $\sim$0.1 change in $f_{\rm SE}$ between 2 and 8 Gyr that *Kepler* cannot.

7. SUMMARY

1. The raw super-Earth fraction of *Kepler* planets is not monotonic in isochrone age: lowest below 1 Gyr (0.31 ± 0.05), highest at 1–2 Gyr (0.48 ± 0.03), flat or declining beyond.
2. The old-age behaviour is set by host demographics and transit sensitivity, not by age; a stratified permutation null shows the confounders alone would produce a young–old difference of the opposite sign ($z = 3.7$ in the combined sample).
3. After matching in mass, metallicity, period and detectability, hosts younger than 1 Gyr have $f_{\rm SE} = 0.31$ against 0.47 ($\Delta = -0.16$, $p = 0.007$), independently in CKS and non-CKS hosts and across fifteen analysis variants; the deficit sits at 1.0–1.5 $R_\oplus$ with a compensating excess at 1.8–2.3 $R_\oplus$. Beyond 2 Gyr the matched difference is -0.02 ± 0.02.
4. An errors-in-variables fit gives a conversion timescale $\tau = 0.4$ Gyr (2σ: 0.15–1.5 Gyr); continued log-linear evolution is only marginally preferred over a static population ($\Delta\chi^2 = 2.7$).
5. A power analysis shows *Kepler* isochrone ages would have detected the Gyr-scale rise claimed in the literature; its absence limits late evolution to $\Delta f_{\rm SE} \lesssim 0.08$ between 2 and 8 Gyr, which does not discriminate between core-powered mass loss and photoevaporation.

DATA AND CODE AVAILABILITY

All inputs are public (GKSPC, DR25 KOI table, CKS VII, Mazeh et al. 2015 rotation periods; mirrors listed in Section 2). The analysis scripts, the derived planet tables with scaled radii, classifications, ages and covariates, and every number quoted in this paper are provided as supplementary material (`radius_age_analysis.zip`).

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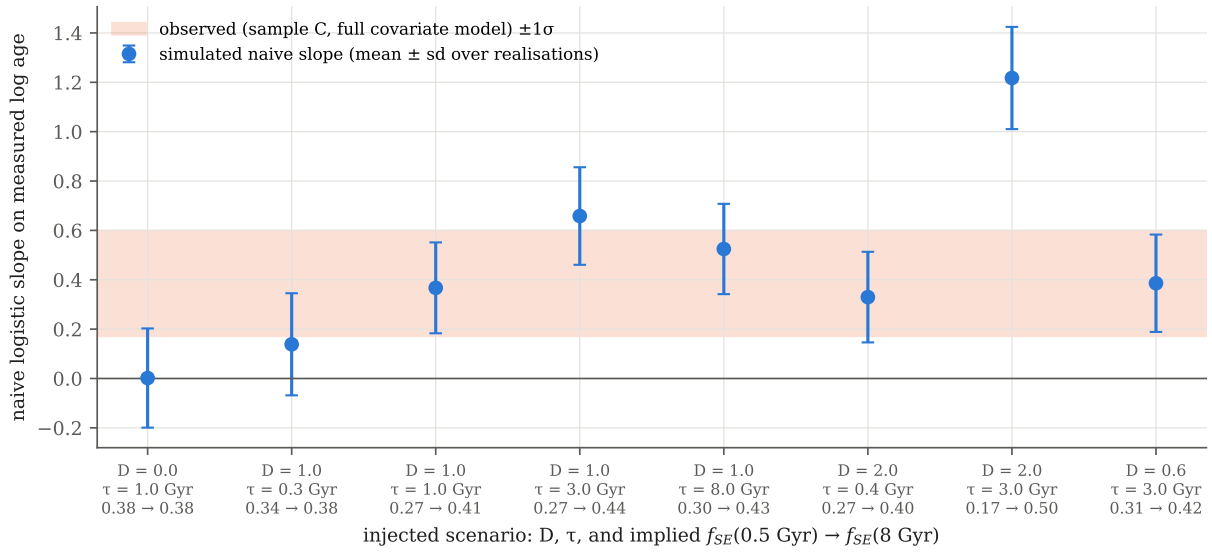


Figure 7. Naive measured-age slopes (mean ± scatter over realisations) produced by injected evolution scenarios in sample C, against the observed value with its 1σ band.

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